WORKED SOLUTION

STEP II 2023, Question 12

Probability and Statistics • Distribution of a Maximum

STEP II 2023, Question 12

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2023 STEP papers (a zip file):

www.ocr.org.uk/Images/707102-2023-past-papers.zip

The key idea

The maximum of several numbers is at most t exactly when *every* one of them is at most t, so the cumulative distribution function of Y is a power of the cumulative distribution function of a single X_i. Everything else follows from working with cumulative distribution functions rather than densities: the median comes from solving $\mathrm{P}(Y \leq m) = \frac{1}{2}$ directly, and the mean comes out neatly from integrating by parts so that the distribution function, not the density, appears in the integral.

Solution

Part (i)

Write $F(t) = \mathrm{P}(X_1 \leq t)$. For $0 \leq t \leq \pi$,

$$F(t) = \int_0^t \tfrac{1}{2} \sin x \, \mathrm{d}x = \tfrac{1}{2}(1 - \cos t).$$

The event $\{Y \leq t\}$ happens exactly when $X_1 \leq t$, $X_2 \leq t$, …, $X_n \leq t$ all happen: if the largest is at most t then so are all the others, and conversely. The X_i are independent and identically distributed, so the probability of this intersection is the product of the individual probabilities:

$$\mathrm{P}(Y \leq t) = \mathrm{P}(X_1 \leq t)\,\mathrm{P}(X_2 \leq t) \cdots \mathrm{P}(X_n \leq t) = [\mathrm{P}(X_1 \leq t)]^n.$$

Hence, writing G for the cumulative distribution function of Y,

$$G(t) = \frac{(1 - \cos t)^n}{2^n} \qquad (0 \leq t \leq \pi),$$

with $G(t) = 0$ for $t < 0$ and $G(t) = 1$ for $t > \pi$. Differentiating by the chain rule, the probability density function of Y is

$$g(t) = \frac{n\,(1 - \cos t)^{n-1} \sin t}{2^n} \quad \text{for } 0 \leq t \leq \pi,$$

and zero otherwise.

Part (ii)

The median $m = m(n)$ satisfies $G(m) = \frac{1}{2}$:

$$\left(\frac{1-\cos m}{2}\right)^n = \frac{1}{2} \iff \frac{1-\cos m}{2} = 2^{-1/n}$$
$$\iff \cos m = 1 - 2^{1-1/n},$$

taking the positive real nth root since $(1-\cos m)/2 \geq 0$. As m must lie in $[0, \pi]$, where cosine is one-to-one,

$$m(n) = \arccos(1 - 2^{1-1/n}).$$

As n increases, $1/n$ decreases, so $2^{1-1/n}$ increases (towards 2) and $1 - 2^{1-1/n}$ decreases (towards -1). Since arccos is a decreasing function, $m(n)$ *increases* with n, and $m(n) \to \arccos(-1) = \pi$ as $n \to \infty$. (For $n = 1$ we get $m(1) = \arccos 0 = \pi/2$, as the symmetry of $\frac{1}{2}\sin x$ about $\pi/2$ demands.) This makes sense: the maximum of more and more values is pushed towards the top of the range.

Part (iii)

By definition $\mu(n) = \int_0^\pi t\, g(t)\, \mathrm{d}t$. Since $g = G'$, integrate by parts with $u = t$ and $\mathrm{d}v = g(t)\,\mathrm{d}t$, $v = G(t)$:

$$\mu(n) = \Big[\, t\, G(t)\Big]_0^\pi - \int_0^\pi G(t)\, \mathrm{d}t$$
$$= \pi \cdot 1 - 0 - \frac{1}{2^n}\int_0^\pi (1-\cos x)^n\, \mathrm{d}x,$$

using $G(\pi) = (\frac{1+1}{2})^n = 1$. This is the required formula.

> **Ask yourself** Why integrate by parts rather than trying to evaluate $\int t\,(1-\cos t)^{n-1}\sin t\,\mathrm{d}t$ directly?

The target formula contains $(1-\cos x)^n$, which is $2^n G(x)$, the *distribution* function rather than the density. Integrating by parts is precisely the move that trades $t\,g(t)$ for $G(t)$, and it removes the awkward factor of t at the same time.

(a) Put $h(x) = \frac{1}{2}(1-\cos x)$, so that $0 \leq h(x) \leq 1$ on $[0, \pi]$, with $0 < h(x) < 1$ for $0 < x < \pi$. The formula reads $\mu(n) = \pi - \int_0^\pi h(x)^n\, \mathrm{d}x$, so

$$\mu(n+1) - \mu(n) = \int_0^\pi (h(x)^n - h(x)^{n+1})\, \mathrm{d}x = \int_0^\pi h(x)^n (1-h(x))\, \mathrm{d}x.$$

On $0 < x < \pi$ the integrand is continuous and strictly positive, because $h(x) > 0$ and $1 - h(x) > 0$ there. So the integral is strictly positive and $\mu(n+1) > \mu(n)$ for every $n \geq 1$: $\mu(n)$ increases with n.

(b) With $n = 2$, expand and use $\cos^2 x = \frac{1}{2}(1+\cos 2x)$:

$$\int_0^\pi (1-\cos x)^2\, \mathrm{d}x = \int_0^\pi \Big(1 - 2\cos x + \tfrac{1}{2} + \tfrac{1}{2}\cos 2x\Big)\mathrm{d}x = \tfrac{3}{2}\pi,$$

since $\cos x$ and $\cos 2x$ both integrate to zero over $[0, \pi]$. Hence

$$\mu(2) = \pi - \tfrac{1}{4} \cdot \tfrac{3}{2}\pi = \frac{5\pi}{8}.$$

From part (ii), $m(2) = \arccos(1 - \sqrt{2})$. Both $5\pi/8$ and $m(2)$ lie in $[0, \pi]$, where cosine is strictly decreasing, so

$$\mu(2) < m(2) \iff \cos\tfrac{5\pi}{8} > \cos m(2) = 1 - \sqrt{2}.$$

Ask yourself How do we get an exact value for $\cos\frac{5\pi}{8}$?

Write $\cos\frac{5\pi}{8} = \cos(\frac{\pi}{2} + \frac{\pi}{8}) = -\sin\frac{\pi}{8}$, and use the half-angle formula $\sin^2\frac{\pi}{8} = \frac{1}{2}(1 - \cos\frac{\pi}{4}) = \frac{2-\sqrt{2}}{4}$. As $\sin\frac{\pi}{8} > 0$,

$$\cos\tfrac{5\pi}{8} = -\tfrac{1}{2}\sqrt{2 - \sqrt{2}}.$$

So we need $-\frac{1}{2}\sqrt{2-\sqrt{2}} > 1 - \sqrt{2}$, that is, $\sqrt{2} - 1 > \frac{1}{2}\sqrt{2-\sqrt{2}}$. Both sides are positive, so this is equivalent to the inequality between their squares, multiplied by 4:

$$\begin{aligned} 4(3 - 2\sqrt{2}) > 2 - \sqrt{2} &\iff 10 > 7\sqrt{2} \\ &\iff 100 > 98, \end{aligned}$$

where the last step squares two positive numbers. This is true, and every step was an equivalence, so $\cos\frac{5\pi}{8} > 1 - \sqrt{2}$ and therefore $\mu(2) < m(2)$.

Checks

- With $n = 1$, $Y = X_1$ is symmetric about $\pi/2$, and indeed $\mu(1) = \pi - \frac{1}{2}\int_0^\pi (1 - \cos x)\,\mathrm{d}x = \pi/2 = m(1)$.
- Numerically, $\mu(2) = 5\pi/8 \approx 1.9635$ and $m(2) = \arccos(1 - \sqrt{2}) \approx 1.9979$, so the mean sits just below the median. A computer simulation of 10^5 maxima for each n from 1 to 7 agreed with both formulas to within 0.01 (for instance $\mu(3) \approx 2.1598$, $m(3) \approx 2.1986$), and both sequences increased.
- The density of Y for $n = 2$ is $\frac{1}{2}(1 - \cos t)\sin t$, which is skewed to the left (a long tail towards 0), and a left-skewed distribution typically has its mean below its median.

Common slips

- Forgetting to justify $\mathrm{P}(Y \le t) = [F(t)]^n$: both the "all are at most t" equivalence and independence are needed.
- Integrating by parts but forgetting the boundary term $\left[tG(t)\right]_0^\pi = \pi$, which is where the leading π comes from.
- Comparing angles via cosines without noting that cosine is *decreasing* on $[0, \pi]$, and so getting the inequality the wrong way round.

Beyond STEP

The maximum of a sample is an example of an *order statistic*, and the trick in part (i) gives the distribution of any sample maximum. The identity $\mathrm{E}[Y] = \pi - \int_0^{\pi} G(t)\,\mathrm{d}t$ is a case of the general tail-integral formula $\mathrm{E}[Y] = \int_0^{\infty} \mathrm{P}(Y > t)\,\mathrm{d}t$ for non-negative random variables, which you will meet early in a university probability course. Both ideas are developed in Grimmett and Welsh, *Probability: An Introduction*.

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